NAIMIRIS A PROFIT DRIVEN SMART FARMING MODEL FOR SCOTCH BONNET PEPPERS

Common Integrated Solution Report

Group R26-IT-103

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Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

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Information about the use of Generative AI and automated assistance in the preparation of this report is provided in Appendix E. We are responsible for the accuracy, originality, citations, code, data, analyses and conclusions. This report includes evidence for the three included individual components; the fourth component is explicitly stated as an evidence gap.

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ABSTRACT

There is a need to make connected decisions regarding land, investment, crop development and plant health when growing scotch bonnet pepper. NAIMIRIS tackles the complexity of how decisions are made around crop health monitoring, crop-health technologies and guidance to farmers via image analysis in the pre-cultivation stage. This is a common report that is created based on three individual theses based upon a design science approach and a focused review of research, agricultural guidance and official technical documentation. The built-in design ties up plot and resource evaluation with an approved farm plan, links observations of the environment and the stage image to a crop cycle, and stores the results of the leaf screening in the same evidence history. The overhead land-cover analysis, two-photo preparation assessment, resource and financial calculations, ESP32 and Firebase monitoring, and stage-specific YOLO detectors are all part of component methods, as well as a seven-category EfficientNet-B0 leaf classifier [1]. The single most important contribution that is integrated is a common contract for identity, units, model versions, uncertainty and controlled recovery. The source documents mention the prototypes of sensing and mobile prediction, and a crop-health preparation workflow that retains the images of 4507 selected images, including 4473 correctly. They are not providing final held out model metrics, a full end-to-end test record or measured farming benefits. The fourth element of the common template that is not determined from the provided evidence is. As a result, the report contains a validated and grounded integration design and bounded prototype account, accompanied by explicit acceptance tests and a progressive validation programme. Any claims of improved yield, profit or unrestricted operational readiness must be validated locally and reviewed by experts and subjected to a field evaluation in the near future.

Keywords: Scotch bonnet pepper, smart farming, decision support system, Internet of Things, crop-health screening

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LIST OF ABBREVIATIONS

AI	Artificial intelligence
AP	Average precision
API	Application programming interface
BBWV2	Broad bean wilt virus 2
C1–C4	Component positions one to four
CNN	Convolutional neural network
DSR	Design science research
EC	Electrical conductivity
EXIF	Exchangeable image file format
FAO	Food and Agriculture Organization
IoT	Internet of Things
IoU	Intersection over union
IPM	Integrated pest management
JSON	JavaScript Object Notation
LKR	Sri Lankan rupee
mAP	Mean average precision
PP2	Progress Presentation 2
RGB	Red green blue
ROI	Return on investment in financial calculations
SDG	Sustainable Development Goal
UTC	Coordinated Universal Time
WCAG	Web Content Accessibility Guidelines

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1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and Context

The decisions made in the Scotch Bonnet pepper cultivation process are connected, such as land-decision, investment-decision, crop care-decision and plant health-decision. A farmer may be able to find land but not have accurate information on preparation costs, soil restrictions, water availability or the number of plants that the available resources can support. Once planted, it is the same farmer who has to interpret changing environmental conditions and the symptoms seen. NAIMIRIS tackles this continuity of decisions by introducing one single system of information for farmers. A Profit Driven Smart Farming Model for Scotch Bonnet Peppers is the title of the proposed purpose for the decision supportive model, not proof of profitability. This common report is based on the individual pre-analysis thesis, the cultivation-guidance thesis and the crop-health thesis.The evidence for this common report is the individual pre-analysis thesis, the cultivation-guidance thesis and the crop-health thesis. The documents outline developmental stages of complementary functions. The first two are primarily focused on Chapters 1-4 and so focus on methods and design. The third contains chapters on implementation and evaluation status. They are read together if they share the same architecture and an account of what has been reported as progress in the prototype, but not a finished field-effectiveness study. Intended crop is Scotch Bonnet (Capsicum chinense). General capsicum and chilli recommendations set out guidelines and background information, but do not automatically determine requirements for all local varieties. In Sri Lanka, the Department of Agriculture has provided guidelines on cultivation and disease management of capsicum [1]. In this project such guidance is considered a reference to be interpreted by the crop and site. The agronomic value must not obscure the crop identity, measurement basis and source of the agronomic value but should remain an unexplained constant in its application. Land evaluation is helpful as a basis to make decisions prior to cultivation. The framework provided by FAO links land variables to a specified use, and distinguishes between current and potential suitability [2]. NAIMIRIS takes that principle one step further and states that, as long as there is no soil, drainage, water or management evidence, apparently open land can only be a

candidate for cultivation. The investment decision can't be made with just a land-cover label. Farm management is also an uncertain process. Prices, yields and costs to produce output vary, and some initial cash might be needed more than a household can raise if the economy looks good. Uncertainty in farm decisions and analysis of uncertainty is discussed in the farm-management material of the FAO [3]. The NAIMIRIS planning design thus distinguishes between physical capacity, the feasibility of resources, seasonal economic return, and when the cash is needed. The calculations corresponding to these are given in this report and the degree of accuracy in the forecasts measured is a task yet to be evaluated. These decisions can be linked together with a common digital record. Accepted farm plan outlines the plot, the way it will be cultivated and the number of plants to be produced. Planting dates, stage observations and sensor history are added when a crop cycle is made. A new health observation, called leaf-image screening, is added to that cycle. Integration is helpful if every subsequent observation carries over the context and constraints of the preceding decision, but instead of presenting three independent predictions to the farmer, it is helpful. The research takes a design science approach. Peffers et al. outline a process for problem identification, design, demonstration and evaluation [4]. In this case, the artefact consists of the common decision-making process, interfaces of the components, and an interpretation for farmers. The validity of the resulting research claim will be dependent on the evidence provided in each of the stages. A design result is a specified interface, a working demonstration is a form of evidence and an independently evaluated farm outcome is a different and stronger form of evidence.

1.2 Stakeholders and User Needs

The intended audience is small-scale growers and potential growers requiring hands-on assistance throughout the growing process. Their requirements differ. A new grower requires assistance to determine if an intended plan is financially feasible and affordable. The experienced grower might primarily require regular past environmental notes, growth and periodic symptoms. They are not personas after they have been completed through an interview study [5] but they are profiles of intended users, obtained from the member reports. Agricultural advisers are secondary users since some of the outputs require interpretation in addition to the measurements

available [6]. Advisers should have access to the source image, the measurement basis, and the missing evidence and recommendation history. A short description of the reasons why a result is provisional can be more helpful than a single suitability score that obscures the reasons. Researchers and maintainers require reproducible inputs, versioned models and error records so that a reported issue is reproducible for investigation. To the integrated system are related stakeholder needs, as shown in Table 1.1. It does not document stakeholder endorsements for requirements or proposed methods of validation.

Table 1.1 Stakeholder needs and validation evidence

Stakeholder	Principal need	Evidence needed to validate the need
Prospective grower	Understand land constraints and initial funding	Plan-comprehension tasks and interviews
Active grower	Interpret stage, sensor and health observations	Crop-cycle tasks on representative farms
Agricultural adviser	Review uncertainty and supporting observations	Structured review of anonymized cases
Research team	Reproduce models and recommendations	Dataset, model and rule-version records
System maintainer	Detect failure and restore service	Monitoring logs and recovery exercises

The application should maintain farmer control. Users agree to the plot and correct resource information and accept advisory actions. The output from machine must not be confused with user confirmation. The farmer might adjust a desired plant count after visiting access ways. The revised count should not be entered silently as an analysis of the original analysis should be saved as an analysis plan revision with a reason, not a new analysis.

There must be clear restrictions on responsible use, too. NIST's AI Risk Management Framework provides a basis for considering reliability, transparency and risk throughout an AI system's lifecycle [7]. That approach is used in the project when making expenditures or crop management decisions that might impact the project. The

interface might be convenient but unsupported diagnoses, hidden missing data and false certainty is considered as a design failure.

1.3 Alignment with the Sustainable Development Goals

The proposed contribution is mainly to SDG 2 (small-scale producer productivity, sustainable agriculture) [8]. Land planning, checks of resources and timely observation may contribute to these ends if they enable the farmer with better decisions. A change in farm productivity or farm income has not yet been measured as part of the project, therefore the alignment is represented as a plausible pathway and not as an impact that has been achieved. Sustainable consumption and production, efficient resource use and sound management of chemicals and waste, are covered in Goal 12 [9]. In NAIMIRIS, this encourages not making unsupported input suggestions, maintaining water and labour limits, and considering the life of sensing equipment. Sustainability doesn't necessarily mean the software is sustainable just by adding sensors or AI. The benefits should be compared with the actual farming practice (including costs and potential unintended benefits). The indicators to evaluate these contributions are developed in Chapter 7. A group-level question is whether the combined workflow enhances the quality of decision making without adding too much weight of equipment, connectivity or maintenance obligations.

1.4 Purpose and Scope of the Common Report

This report gives an overview of the common problem, architecture, integration needs, implementation evidence and the evaluation plan of the three components provided. Follows the Common Report Content structure of 8 chapters. The typography, spacing, margins and pagination are governed by a separate formatting standard. Only the subject matter, experimental findings and numerical results of the example group report were used to understand the presentation conventions only; it is not evidence for NAIMIRIS.

The regular template provides for four items. Unfortunately, only three member reports were provided for this synthesis and the general descriptions of the wider projects do not provide one consistent fourth component. For this reason, the fourth-

component position was kept in both the evidence gaps in Sections 2.3 and 3.6. A fourth owner, technical implementation, result or contribution is not made up. This boundary also reflects the fact that the report does not claim that it provides a full four-component solution.

The main part of the report is a coherent integration specification, based on the member documents. Reconciles common identifiers, sensor differences, image-task boundaries and evidence states. It also defines conclusions that can be drawn and areas that should be inspected in the repository, tested during execution or evaluated in the field. Details of individual components are given only when needed to explain integration or reproducibility or interpretation of results.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall Problem Definition

The challenge is that the information and data used to make decisions for all aspects of Scotch Bonnet production may be disconnected and unequal. The appearance of the land, resources of the farmer and environmental readings or symptoms of the leaves answer different questions. If there is no common context, a farmer might consider one positive observation as enough evidence to make an investment or crop management decision. On the other hand, there can be key limitations that are missed between planning and daily cultivation. NAIMIRIS explores what needs to be done to maintain them while also making the information usable. A system should outline the observations that justify a recommendation, any assumptions that are not confirmed and the appropriate action that should follow. The challenge is therefore not just for the image-classification score to be high. It involves gathering appropriate inputs, matching them to the farm and time, understanding the constraints and expressing a course of action, which the farmer can discuss. It is illustrated by a representative case. Open ground may be part of a plot and can be represented with a preliminary area-based plant count. If the capacity of water is reduced, then the accepted count should be changed. When cultivated, a sensor might inform the dry condition; however, an old sensor timestamp or an outdated calibration must not make an automatic conclusion of irrigation. The score for the disease may then be added to the curled leaf, but this score will not confirm that the moisture score was the root cause of the curled leaf. A regular system should not be interpreted as diagnosis, it should only hold all three observations. The overall research question is thus: What can be done to have land and resource planning, stage-aware environmental guidance and image-based crop-health screening all be part of a single traceable decision-support workflow for the intended crop, without losing the uncertainty and human review?

2.2 Root Causes and Overall Research Gap

2.2.1 Different observations have different meanings

Land-cover regions are labeled using an overhead segmentation model, visible preparation conditions are recognized by a ground-image model, visible crop organs

are detected by an object detector and an image category is classified by a leaf classifier. No interchangeability of these outputs. All four task types are used in the member reports, thus their interfaces are as crucial as their algorithms per se. Both LoveDA and OpenEarthMap offer a land cover map composed of two domains: urban and rural, and geographically diverse scenes, respectively [10] and [11]. They are relevant only if there is annotated land-cover material available. Both do not define agronomic suitability of a plot of Sri Lankan Scotch Bonnet. This separation results in a project-specific mapping layer with ambiguous classes and further evidence required before recommending cultivation. U-Net is a well-established encoder-decoder segmentation network [12]. It offers a valid contender for the comparative land-model studies, but does not reveal the actual structure of the land model employed in the given pre-analysis work. The common report thus specifies the land model as the one that has been documented in terms of its task and input/output contract. The selection of architecture is still a fact at the checkpoint level to be determined from configuration and weights. Crop-health task is more precisely defined. The seven-category classifier [13] is provided by EfficientNet as its underlying architecture. Previous studies on plant diseases have shown the technical possibility of learning image classification [14]. Those studies inspire the approach but do not convey their benchmark scores to NAIMIRIS. The contribution here is about a consistent flow of work and its assessment in the context intended

2.2.2 Missing and uncertain evidence can propagate

A score might be accurate but be based on the wrong data. The reason for not being able to equate the confidence values with the frequency of correct predictions is the subject of neural-network calibration research [15]. At system level, this does not mean that if a leaf classification score is high, that the system should ignore an unsupported crop or image type warning. Similarly, soil evidence should not be replaced by a high confidence land label.

The concept of messiness should be explicitly represented. The absence of an electric conductivity reading does not mean the same as zero. A failed image does not mean that the image does not contain a detectable fruit. Crop cycle with an unconfirmed stage is different from crop cycle confirmed as germination. If these conditions are

combined within the common data model, the guidance will be unreliable even if there are no problems with each component of the common data model returning a syntactically valid response.

2.2.3 Shared context is absent from isolated tools

Independent tools can be used to assist with individual tasks, but the farmer will still need to balance the output from those tools. NAIMIRIS fills the gap of the design of a crop-specific workflow that allows an observation to provide information for a decision, along with the plot, the cycle, the timestamp, the model or calibration version. It is an integration contribution; it is NOT an assertion that there do not exist agricultural decision-support systems already. Table 2.1 summarizes related technical approaches, and the extra integration effort that will need to be done. The table can be taken as a conceptual comparison only, and does not compare the capabilities of commercial products or experimental scores.

Table 2.1 Related approaches and implications for integration

Approach	Established role	NAIMIRIS integration requirement
Land-cover segmentation [10], [11]	Pixel-level cover interpretation	Preserve unknown area and require soil evidence
YOLO object detection [16], [17]	Locate and classify visible objects	Retain image scope, stage and model label map
Efficient Net classification [13]	Image-category prediction	Preserve seven-class mapping and preprocessing
ESP32 sensing [18]	Acquire and transmit device observations	Add calibration, units and farm assignment
Firebase rules [19]	Control database access and validation	Apply ownership consistently across records
FAO land and farm frameworks [2], [3]	Structure suitability and resource decisions	Make assumptions and constraint calculations explicit

Official YOLOv8 and YOLO11 documentation categorizes the model families and the tasks that can be performed with the model [16]. This is significant because there

are various types of detectors that are explained in the guidance report for emergence of seedlings and fruiting [17]. The documentation of the ESP32 platform outlines the connectivity options of the hardware platform [18] and the documentation of Firebase outlines how access and validation can be done with rules [19]. These sources do not dictate what technology to use, but rather that the NAIMIRIS deployment incorporates some of the required controls.

2.2.4 Review method and source selection

Targeting searches, checking of original research proceedings, official documentation, and institutional agricultural guidance and dataset records were used to complete the external review on 11 October 2026. The searches included land suitability, farm uncertainty, datasets of land-cover, EfficientNet, confidence calibration, YOLO variants, telemetry and pepper direction and responsible software operation. Sources were kept where they helped make a decision for a particular design or evaluation.

This was a concisely written narrative review and not a systematic review. No screening counts are invented and no claims are made of complete coverage of the literature. The individual reports' references did not reproduce verbatim: irrelevant, duplicate or inconsistent references were replaced by one group bibliography. The three member reports are the source of project specific implementation claims, published research is used for background and method justification.

2.3 Sub problem and Component Mapping

There are three different sub problems that are related. The first one is related to feasibility of a proposed farm plan prior to investment. The second is what to do once the crop is planted and the environmental factors have been planted. The third relates to the inspection of leaf symptoms and the appropriate response to be advised. Substantiating evidence to support the fourth position outlined in the template is not possible from the available member evidence.

The boundaries of the responsibility are defined in tables 2.2. The fourth position is not visible if it is not kept on display then an incomplete source set can be shown as a verified four-component system.

Table 2.2 Sub problems and component ownership

Position	Sub problem and component	Owner and evidence
C1	Pre-analysis and smart decision support	Kumarage K J B H, IT22271464
C2	Technology-assisted A-to-Z farming guidance	Savinda R.M.D, IT22124630 [20]
C3	AI-driven pest monitoring and control through leaf screening	Gunawardana T.D.A, IT22212740
C4	Fourth subproblem not established	No member report supplied; excluded from implementation claims

C1 provides C2 with an approved plan and its constraints. A C3 health observation can be accompanied with crop-cycle and stage background provided by C2. C3 may initiate an adviser review or inspection, but it cannot alter the initial financial projection, replace the crop stage, or provide a verified pathogen diagnosis. A changed plan can be supported by later observed costs, losses, and harvested output if they are clearly documented. This feedback is not a closed-loop result that has already been measured; rather, it is a defined integration method.

2.4 Overall Aim and Group Objectives

The overall goal is to create and validate an integrated decision-support process for Scotch Bonnet cultivation that uses shared, traceable evidence to link stage-aware monitoring, bounded crop-health assessment, and practical pre-cultivation planning.

The goals in Table 2.3 are structured so that software and field achievement can be distinguished from documentary achievement. An interface-definition goal may be met by a finished design, but end-to-end implementation may not be confirmed.

Table 2.3 Group objectives and evaluation links

ID	Group objective	Required evaluation evidence
G1	Define one crop-cycle workflow across C1 to C3	Architecture, ownership and dependency review
G2	Preserve identity, units, time and provenance between components	Contract and invalid-input tests
G3	Connect a feasible plan to stage-aware monitoring	Versioned plan-to-cycle scenario

ID	Group objective	Required evaluation evidence
G4	Integrate leaf screening with faithful labels and bounded advice	Preprocessing, mapping and guidance tests
G5	Handle missing, stale and failed evidence consistently	Failure-injection and recovery records
G6	Assess usability, predictive quality and farming value separately	Held-out metrics, user tasks and prospective field data

There isn't a single definition for the entire AI accuracy percentage. Each of the system's functions—segmentation, detection, classification, and deterministic computations—has a distinct ground truth. Instead, group evaluation enquires as to whether the workflow accurately transmits evidence and supports a suitable judgement. Predictive metrics are interpreted in conjunction with user and operational outcomes while staying task-specific.

2.5 Scope Assumptions and Constraints

Map or image-based plot input, precisely two guided preparation photos, resource information entered by farmers, environmental observations, crop-stage picture analysis, and seven-category leaf screening are all included in the supported scope. It's an advisory system. The defined scope does not include autonomous irrigation, pesticide application, laboratory pathogen identification, validated market execution, and a finished fourth module.

The concept is predicated on farmers being able to identify their plots, upload images that might be used, and supply basic resource information. It also presupposes suitable device calibration, local evaluation of crop-management regulations, and legal access to training data and imagery. Instead of being regarded as enduring characteristics of the user base, these presumptions need to be checked.

Unequal sensor coverage is described in the reports. Soil temperature, relative humidity, air temperature, and soil moisture are all specified in C1. Electrical conductivity is also included in C2. As a result, a defined sensor-capability set is supported by the integrated schema. If a device doesn't have EC, it's optional, and any rule that needs it won't work until there's a valid reading with a recognised foundation.

Limited model-training sessions, missing final evaluation artefacts, inconsistent connection, and inadequate local datasets are examples of resource and evidence restrictions. Resumable checkpoints are motivated by a limited training environment, however incorporating test data into training is not justified. A temporary financial assessment, not a fictitious monthly history, is motivated by an unavailable price series. Implementation and reporting are governed by these boundaries.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated Solution Overview

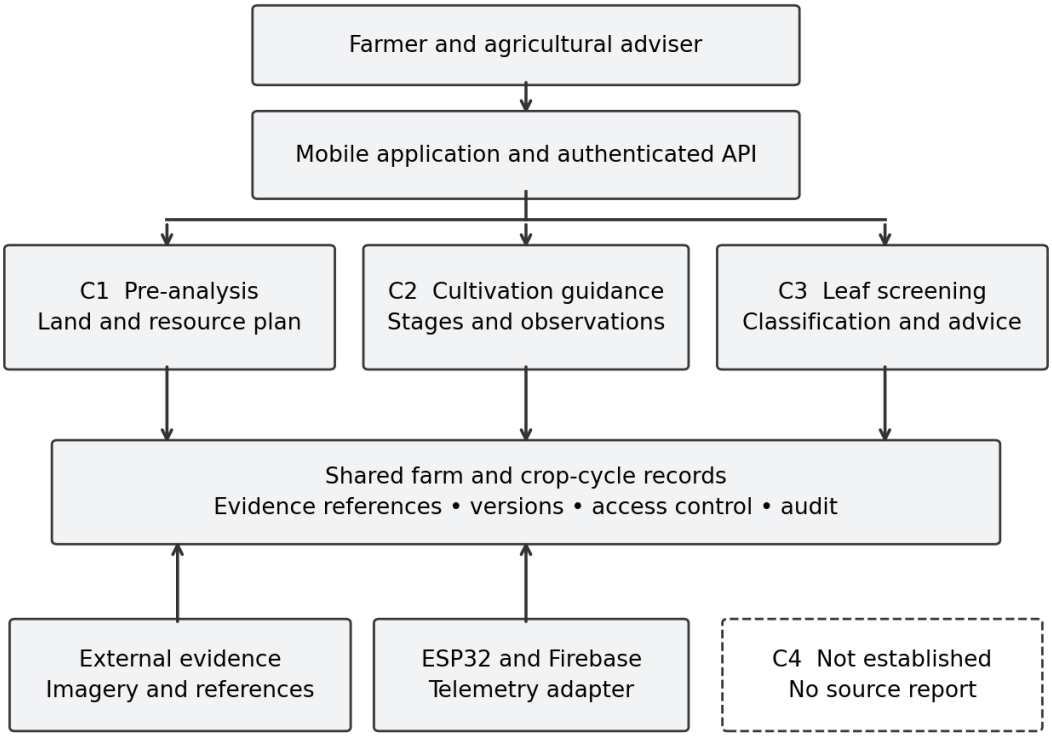
A farm, a plot, and a crop cycle form the foundation of NAIMIRIS. Using land and resource evidence, the user first drafts a plan. The plan becomes the crop cycle's reference once it is approved. That cycle is linked to later sensor and image observations. In addition to allowing the farmer to document what transpired next, recommendations retain the observations that led to them.

While maintaining ownership boundaries, this structure prevents the same contextual information from being entered repeatedly. Planning calculations are still handled by C1, stage and environmental interpretation by C2, and the leaf-classification contract and associated management material by C3. Identity, storage, evidence references, and audit records are all managed via shared services. It is not necessary for the models to share a neural architecture in order to have a shared user interface.

The common system makes a distinction between assumed, inferred, and observed values. Agricultural references have sources and review dates; projections have model versions; farmer entries have authors and edit histories; and sensor measurements have calibration metadata. Realized transactions and financial assumptions are kept apart in storage. Tracing an advisory output back to its inputs is based on this distinction.

3.2 High Level Architecture

The logical integration design derived from the member reports is shown in Figure 3.1. It is not a screenshot of a confirmed production deployment; rather, it is an architecture specification. The common evidence store, component responsibilities, application border, and external sources are all purposefully visible.



Logical design; deployed service configuration is not independently verified.

Figure 3.1 Logical architecture for the three evidenced components

Plot selections and photos are accepted by the smartphone application, which then displays the results. The request is authenticated, ownership is verified, and work is dispatched via the application programming interface (API). In order to save the user from having to restart after a brief network outage, longer-running image and simulation jobs might return job identifiers. Even if the mobile interface makes initial tests, the backend is still in charge of validation.

The reported ESP32 and Firebase pathway is connected to the standard observation format via a telemetry adapter. This pathway is referred to as prototype progress by C2 [20]. Before being referred to be deployed infrastructure, the planned long-term storage and common API layer must be verified against the real repository. Future transport modifications can be made without altering the significance of a soil-moisture observation by separating the adaptor from decision logic.

The storage design makes a distinction between decisions, derived analysis, and raw evidence. A suggestion is a decision-support record that refers to one or more studies; an image is an asset; and a model run is an analysis of that asset. A previously released

report shouldn't be silently changed by deleting or substituting a model run. A new model version produces a new outcome with a discernible connection to the original.

The border for the fourth component is displayed as unavailable. The primary workflow cannot wait endlessly for information from an unidentified service. It is possible to enter crop-reference or external pricing records with provenance without asserting the implementation of a marketplace or post-harvest module.

3.3 Component 1 Summary

3.3.1 Owner inputs and methods

Pre-Analysis and Smart Decision Support are owned by Kumarage K J B H, IT22271464. A plot border, overhead photos, precisely two ground photos, soil and environmental proof, a budget, labour, water, and equipment data are some of its inputs. The ground model detects obvious preparation conditions, while the overhead model offers land-cover evidence. Suitability, resource limitations, and financial possibilities are assessed by a different decision layer.

The area-calculation technique and coordinate system of the plot geometry must be maintained. Geographic degrees cannot be utilised directly as square meters; instead, GeoJSON employs longitude-latitude coordinates in WGS 84 [21]. As a result, the design records an appropriate geodesic or projected-area calculation and stores the original boundary. Survey-quality area estimates cannot be automatically supported by uploaded non-georeferenced images, which necessitates additional assumptions.

Agriculture, open or barren terrain, buildings, roads, water, trees, and other and unknown categories are all preserved in the land-class mapping. Rangeland on OpenEarthMap is not quietly transformed into ready-to-use agricultural land. Rather than being dispersed among confident classes, unknown pixels stay in the denominator. This keeps a plot from appearing more plausible than the evidence warrants due to missing coverage.

3.3.2 Planning outputs and calculations

Verified candidate area and spacing serve as the foundation for the provisional physical capacity. Capacity is floor (A × e / (r × s)) if A is useable square meters, e is

the residual layout factor, r is row spacing, and s is plant spacing. If access space has already been subtracted from A, the factor e cannot take that space away again. A subsequent layout check can be necessary if the plot form is irregular.

The final number of plants that can be grown is limited by manpower, water, land, cultivation method, and budget. N = min(N_land, N_method, N_budget, N_labor, N_water) is a simplified separable computation. The same season and management assumptions must be used to calculate each bound. Instead of assuming that a single closed-form minimum resolves all cases, the engine must test candidate plans when fixed costs and resource interactions render the bounds non-separable.

First-season cash requirements and seasonal economic costs are distinguished by the financial design. While asset use or depreciation falls under economic cost, new capital purchases have an impact on cash flow. Planted count, survival, yield basis, and approved grade are used to calculate salable production. Revenue is based on the harvest period and the appropriate selling channel. A farm-gate pricing does not always follow from a retail listing.

A plan revision with the suggested count, binding limitations, preparation tasks, cost assumptions, appropriateness status, and unresolved evidence is the result. A tentative assessment results from missing crucial soil evidence. Future production and profitability are not guarantyd by this output, which is used as the input for cultivation registration.

3.4 Component 2 Summary

3.4.1 Owner inputs and methods

Technology-Assisted A-to-Z Farming Guidance is owned by Savinda R.M.D., IT22124630 [20]. Crop-cycle data, sensor observations, and checkpoint photos from five stages—germination, seedling establishment, vegetative development and branching, blooming and pollination, and fruiting and ripening—are combined in this component. Stage dates can be adjusted by observation and are used as scheduling tools.

ESP32-to-Firebase sensing, YOLOv8n seedling identification, a YOLO-based flowering detector, and YOLO11s fruit detection are all included in the member report.

The precise version of the flowering model is unknown. The final implementation does not adopt an earlier ResNet-18 stage-classification model. By making these distinctions, historical design alternatives are kept from turning into statements about the deployed pipeline that are in conflict with one another.

Soil moisture, soil temperature, ambient temperature, relative humidity, and, if available, EC are all included in telemetry. Every variable needs its own foundation. Different quantities include the proportion of field capacity, volumetric water content, and raw moisture percentage. EC also requires a documented unit and technique of measurement. An unqualified device value cannot be compared by the system to a threshold set for a different basis.

3.4.2 Guidance outputs and interpretation

Stage context, visible counts, environmental state, and recommended observations or actions are all produced by C2. A valid image result and a suitable time range of valid sensor data are linked by the rules. While being left out of a current recommendation, an outdated reading might be displayed as history. When a result is dependent on a proven stage or a provisional stage estimate, the user should be able to see it.

Fruit counts that are visible are measures at the picture level. If the denominator is positive and the class definitions are verified from the checkpoint, the ripening fraction is N_ripening / (N_unripe + N_ripening). It is not proof that all fruit is immature because no detections result in an unquantifiable fraction. Comparisons between dates are limited by occluded fruit and altered camera framing.

Persistence checks are suggested by the source design to lessen recurring notifications. An alert can be opened by repeated out-of-range aggregates and closed by repeated acceptable readings. These are not predefined crop-response constants; rather, they are changeable engineering rules that need to be evaluated. Instead of being mislabeled as crop stress, device defects must generate device warnings.

An continuing record of how the planted crop grows within the parameters of its approved plan is the primary contribution to integration. Inspection can be informed by a stage shift or new observation, but the initial investment assumptions shouldn't be revised. A new plan or projection revision should include updated expectations.

3.5 Component 3 Summary

3.5.1 Owner inputs and model contract

The AI-Driven Pest Monitoring and Control System is owned by Gunawardana T.D.A., IT22212740. Its implemented task is to classify leaf images using PyTorch's EfficientNet-B0. The current model does not locate insects, count pests, or run a control device, despite the general component label. It designates the healthy category or one of six disease-related groups.

A seven-output classifier with a fixed index mapping is described in the report. The Exchangeable Image File Format (EXIF) orientation is corrected, the image is converted to red, green, and blue (RGB), and it is fitted into a 224 by 224 square with centred grey padding. The training transformation must be replicated in the deployment. A model comparison could be rendered invalid by a different crop-based transform that eliminates pertinent leaf content.

The class contract is listed in Table 3.1. Descriptive text may be translated via the interface, however output indices cannot be silently renamed or rearranged alphabetically.

Table 3.1 Stable leaf classifier mapping

Index	Model key	Interpretation in the application
0	healthy	Healthy-looking submitted leaf
1	bacterial_spot	Possible bacterial spot
2	cercospora_leaf_spot	Possible Cercospora leaf spot
3	leaf_curl	Suspected leaf curl condition
4	powdery_mildew	Possible powdery mildew
5	anthracnose	Suspected anthracnose
6	bbwv2	Suspected broad bean wilt virus 2 infection

3.5.2 Dataset and advice boundaries

A controlled BBWV2 collection, chosen COLD pictures, and public chilli datasets are identified in the crop-health study. Chilli-image provenance is provided by the COLD record [22]. Laboratory photography of Capsicum annuum is specifically described in

the BBWV2 dataset [23]. Renaming an application result for Nai Miris does not prove that the pictures accurately depict Scotch Bonnet conditions in the area. Thus, a key evaluating concern is domain transfer.

The classifier and the knowledge base are distinct entities. University extension materials [24], powdery-mildew data on UC IPM [25], and more general descriptions of pepper diseases on UF/IFAS [26] can all be used to inform bacterial-spot guidelines. Reviewed guidelines are informed by these sources; recommendations for foreign chemicals are not automatically transferred. Aphid transmission within the appropriate genus is described by ICTV, but BBWV2 maintains its virus meaning [27]. A chemical remedy or pesticide dosage cannot be established by a categorisation score.

A stable category, scores, model metadata, and an uncertainty state are all returned by the component. A healthy-looking outcome is solely applicable to the appearance that was presented. A closed-set classifier may give an irrelevant image a high score, therefore low-confidence and unsupported-image behaviour need to be assessed independently. The severity and damaged area are not estimated by the existing seven-output model.

3.6 Component 4 Summary

A fourth individual report was not available. While the other reports provided concentrate on cultivation direction and crop-health screening, the pre-analysis report discusses marketplace, price intelligence, and post-harvest functions within its larger project context. These descriptions don't specify what has been implemented, who owns it, or which one-fourth of the final common report should be included.

As a result, the design reserves an extension boundary without giving it a service or outcome. The same farm and crop-cycle IDs, access constraints, evidence versioning, and evaluation discipline would be required for a future confirmed component. These records require the commodity, grade, channel, location, and date if it offers prices. Its contract must be based on the member's actual work if it serves a different purpose. Here, there is no assertion of a choice between these options.

3.7 End to End User and Data Flow

An authorized plot and registration are the first steps in the typical design process. After gathering the boundary, pictures, and resource data, C1 either provides an approved plan or a tentative outcome with incomplete proof. When a plan is accepted, an unchangeable reference is created. After that, C2 receives observations for the appropriate plot and device and registers a crop cycle against that reference.

The user submits an image and receives stage-specific observable detections at a due checkpoint. A separate C3 request processes the leaf image under its own preprocessing and class contract if a leaf symptom needs to be examined. Each result has its own timestamps and restrictions when it appears in the crop history. Their results are not combined into an unsubstantiated overall crop-health percentage based on a shared history.

The primary decision gates are depicted in Figure 3.2. The user is returned to the pertinent evidence-collection phase when information is incomplete. This makes the subsequent action particular while preserving finished work.

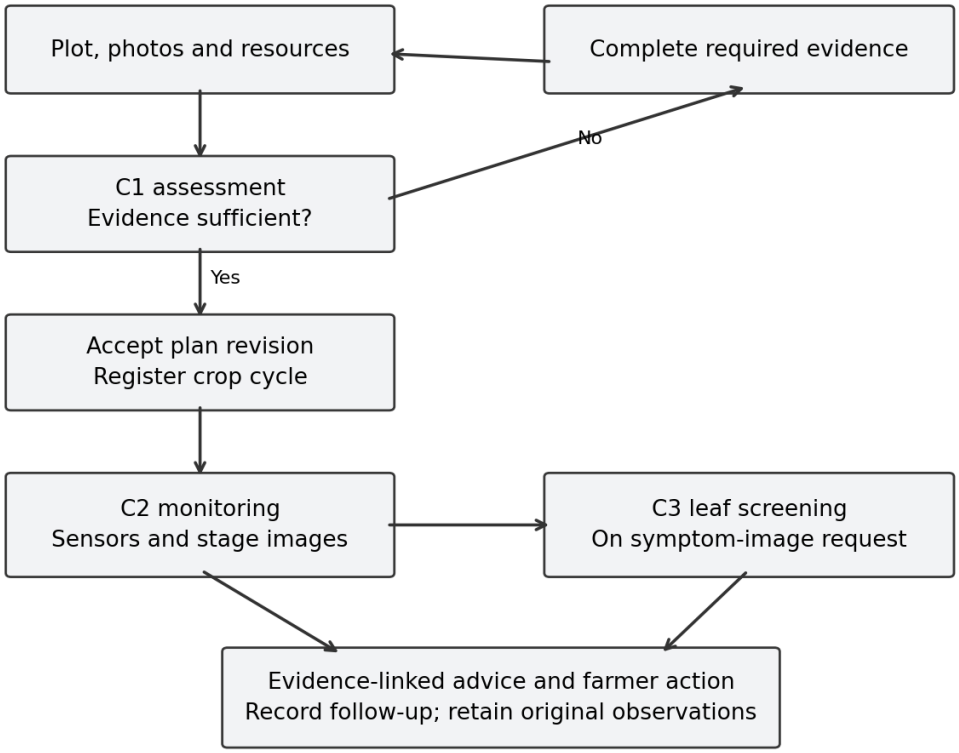


Figure 3.2 Integrated workflow and evidence gates

A revised farmer action is documented independently of the recommendation, such as lowering the number of plants or requesting an expert inspection. The same cycle can be linked to subsequent results. This eliminates the presumption that all recommendations were followed and allows future comparisons between a recommendation, the action actually taken, and the observed consequence.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration Strategy

The integration approach is incremental and contract-first. Before connecting component screens, it fixes the meaning of shared records. The storage descriptions and route prefixes used in the member reports are similar yet inconsistent. As a result, this report treats exact deployed endpoints as a repository-verification item while defining a shared logical contract. The suggested pathways in Appendix A are not assertions of production availability but rather examples for reconciliation.

Identity is the first integration border. Every crop cycle, plan, picture, and observation is part of an approved farm context. The second is proof meaning: immutable versions, timestamps, units, and status values are all agreed upon by the system. Workflow is the third: a plan can be approved, a cycle can be started, an observation can be examined, and an advised action may be noted. A whole farming journey should only be viewed as an end-to-end test once those boundaries have been set.

Connecting C1 plan output to a C2 crop-cycle fixture, followed by telemetry, stage images, and C3 screening, is a workable order. At every stage, failure cases are presented. Both a successful live sensor display and a successful leaf prediction cannot hide an inaccurate plot assignment or an unauthorised image reference. For each boundary, integration activity should document the producer, consumer, schema version, and acceptance fixture.

Compatibility considerations must be made explicitly for version updates. If older customers safely disregard the optional EC measurement and no rule considers its absence as zero, then adding it is backward compatible. Adding an optional display field is not the same as altering a moisture basis or class index. These modifications necessitate a regression fixture, migration procedures, and a new schema or model contract.

4.2 Interfaces Data Contracts and Shared Models

4.2.1 Common identifiers and provenance

Schema_version, request_id, farm_id, plot_id, crop_cycle_id where applicable, observed_at, recorded_at, and evidence references are all included in the suggested

common envelope. Instead of being trusted from an arbitrary request field, the actor identification is determined from authenticated context. The asset checksum, model identification, preprocessing version, and label-map version are also recorded in an image analysis.

The core records are defined in Table 4.1. It is not necessary to use a specific database engine to maintain the meaning of these logical entities, which can be implemented in other repositories.

Table 4.1 Shared records and version boundaries

Record	Essential content	Version rule
Farm and plot	Owner, boundary, area and access permissions	Boundary edits create a new plot revision
Plan	Count, method, resources, costs and limitations	Accepted revisions remain reproducible
Crop cycle	Plot, accepted plan, dates and stage	Stage corrections retain actor and reason
Observation	Device or image, time, units and quality	Raw observations remain distinguishable
Analysis	Input references, model and output contract	New inference creates a new analysis
Recommendation	Rule version, evidence, wording and status	Advice updates preserve earlier issued content
Action and outcome	Farmer response and observed follow-up	Recorded separately from machine advice

When the farm condition was recorded is described by an observation time. The time it was received by the server is described by a recorded time. When a device uploads buffered readings, the distinction is significant. While storage employs explicit time zone-aware timestamps, local displays may utilise Sri Lankan time. If a later arrival's catch time is older, it shouldn't automatically become the most current farm observation.

Evidence classifications are also distinguished by provenance. Measurement, model prediction, farmer input, sourced reference, proxy, and assumption are the suggested values. The identifiers of the supporting inputs are retained for each downstream

decision. This prevents a proxy pricing from being represented as a recorded sale or a model forecast as a sensor measurement.

4.2.2 Telemetry and image contracts

Device identification, measurement name, value, unit, basis, calibration identifier, observation time, and quality status are all necessary for sensor records. The measures that are available are declared in a device-capability record. When measurements are invalid or absent, the numerical value stays null and the observation status describes the absence. The application must clearly indicate its age even though it may show the last legitimate reading.

The intended task, image asset, crop-cycle context, and capture time, if known, are all included in image requests. Geometry alignment is also required for C1 overhead images. C2 requires the pertinent checkpoint and detector context. The leaf-classification preprocessing contract is required by C3. The same image dimensions can indicate essentially various jobs, so a generic image-analysis endpoint that deduces the task only from a filename is inadequate.

Model_state and config are included in the C3 export that is detailed in the member report. The loading code needs to reconstruct a seven-output model and validate that structure. a pest_detector file.A YOLO checkpoint is not implied by pt. Instead of supplying predictions from an incorrect or uninitialised architecture, the startup step must explicitly fail if weights or metadata are incompatible.

4.2.3 Status values and error responsibilities

The terms VALID, PROVISIONAL, STALE, INVALID, UNCERTAIN, and UNAVAILABLE are distinguished by a common status vocabulary; VALID indicates that the pertinent contract checks were successful rather than that a biological conclusion is certain. An incomplete decision is identified with PROVISIONAL. The interpretation policy for a model outcome is subject to UNCERTAIN. UNAVAILABLE denotes the inability of the service or necessary object to produce a result.

Input validation and output information are owned by the producing service. The consuming service verifies the necessary proof and compatibility. Request correlation

and user-visible progress are owned by the orchestration layer. Although the interface has a clear presentation, a backend validation rule cannot be weakened by it. These responsibilities are expanded with exemplary schemas and faults in Appendix A.

4.3 Shared Nonfunctional Requirements

4.3.1 Security and privacy

Financial documents, photos, and farm coordinates all require regular access restriction. One of the main API risks identified by OWASP is object-level authorization [28]. Having a plot or picture identifier is not enough for NAIMIRIS; each record lookup must verify the caller's connection to the farm or shared resource. As a result, cross-farm requests using otherwise legitimate credentials are part of the evaluation plan.

Encrypted transmission, limited retention, validated uploads, restricted image access, and protected device credentials are some of the suggested measures. Images shouldn't be made public just because an analysis is finished. Sensitive payloads should not be routinely copied from logs; instead, the request and error context should be captured. Access limitations must be maintained until the user consciously shares a report that they have requested.

4.3.2 Reliability and performance

Only when the observed interval is evident does a response-time target have any significance. Different delays are caused by upload, queueing, preprocessing, inference, rules, and response rendering. Although a three-second aspiration is mentioned in the source guideline report, there is no comprehensive latency series to support treating it as attained [20]. On a certain device, network, and backend setup, the group should report the median and upper-percentile delay.

Predictable failure is a component of reliability. A duplicate telemetry event shouldn't generate a new observation, a timeout shouldn't attach an outdated analysis to a fresh image, and a missing model should return an unavailable result. When independent work is successful, partial results may still be available, but the application must specify what is lacking. Without monitoring data, no numerical availability claim is made.

4.3.3 Usability accessibility and maintainability

Familiar units, succinct explanations, and obvious corrective paths should all be used in the farmer interface. Although colour might help with status awareness, it shouldn't be the primary signal. Accessibility guidelines for intelligible controls, error communication, and interactivity are provided by WCAG 2.2 [29]. Using specific principles is not a claim of audited conformance; rather, it is a design purpose.

Maintaining the separation of model inference, farming rules, and display phrasing is essential for maintainability. After that, the team can replace a model without subtly altering the rule interpretation or amend a reviewed guideline entry without altering trained weights. Compatible model, schema, and advice versions should be identified with each release. For this aim, a modest set of reproducible fixtures is more beneficial than unrecorded manual demonstrations.

4.4 Integrated Workflow and Exception Handling

4.4.1 Plan acceptance and monitoring

C1 first computes candidate land information and verifies the quality of the input. The user verifies the resources and conditions for preparation. Assumptions and gaps in the evidence are identified via suitability and financial outputs. Only after examining those restrictions does the farmer approve a plan amendment. C2 uses the approved revision, which includes the intended technique and count, to record a crop cycle.

Image inference is unrelated to sensor intake. Telemetry shouldn't be discarded by a brief image-service failure, and a telemetry outage shouldn't keep a user from seeing a clearly labelled historical image result. Until both sources are usable, recommendations that call for both remain tentative. Figure 4.1 maintains a distinct channel for user correction and displays the statuses for an analysis request.

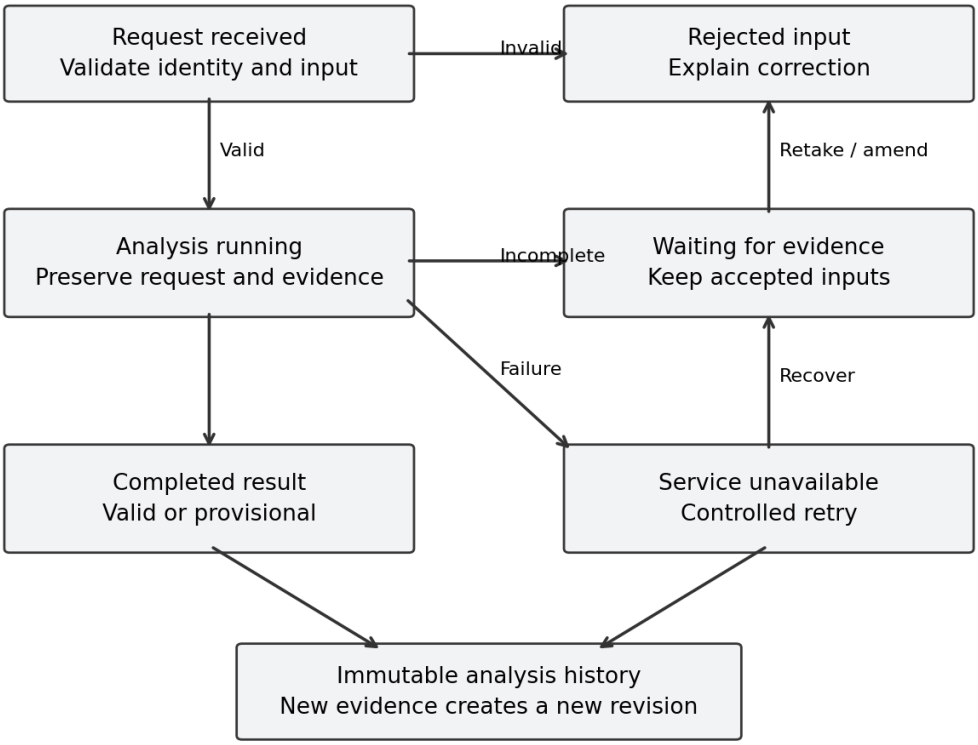


Figure 4.1 Analysis states and controlled recovery

4.4.2 Conflicting recommendations

Evidence and scope should be used to settle disputes. Both a low moisture reading and a leaf prognosis that appears healthy can be legitimate observations. Neither demonstrates that the other is incorrect. The leaf screening outcome and environmental status should be separated in the user-facing output. The device notice supersedes any agronomic interpretation based on that value if the sensor foundation is invalid.

The yield assumption in the financial model shouldn't be automatically lowered by a disease screening result. A supported loss estimate, an observed result, or a clear user assumption are necessary for such a change. A scenario comparison may be provided by the system, but it must remain hypothetical. This keeps an ambiguous classification from turning into a precise claim for monetary loss.

4.4.3 Failure and recovery policy

Table 4.2 summarizes system-level failure cases. Each recovery keeps the original evidence and creates a new attempt or revision where necessary.

Table 4.2 Failure handling across components

Condition	Required response	Preserved evidence
Invalid plot boundary	Reject analysis and request correction	Original boundary and validation reason
One poor ground photo	Request replacement of that photo	Other accepted photograph
Missing required soil evidence	Return provisional suitability	Valid land and resource observations
Stale or uncelebrated telemetry	Suppress dependent guidance	Last valid time and quality reason
No fruit detections	Mark ripening fraction inestimable	Image and detector output
Wrong or missing classifier bundle	Mark service unavailable	Request and startup error reference
Uncertain leaf result	Request better evidence or expert review	Scores and model version
Retry after timeout	Reuse request identity safely	Existing job and result association
Undefined fourth component	Continue supported C1 to C3 flow	Explicit scope limitation

4.5 Team Contribution and Dependency Matrix

The provided member reports are the source of the contribution matrix in Table 4.3. It assigns accountability for the technical and intellectual scope that each member describes. It does not determine individual commit counts or assign percentages. Corroborating repository and project records are necessary for shared implementation claims.

Table 4.3 Member contributions and integration dependencies

Member	Documented responsibility	Shared dependency and evidence needed
Kumarage K J B H	Land evidence, suitability, resources and financial design	Plan contract; model configuration and calculation tests
Savinda R.M.D	Sensor integration, stages, detectors and cultivation guidance [20]	Crop-cycle contract; telemetry and detector logs

Member	Documented responsibility	Shared dependency and evidence needed
Gunawardana T.D.A	Dataset workflow, classifier, class mapping and health guidance	Image contract; export parity and guidance tests
Shared work	Application context and proposed integration in this report	Repository history, review records and end-to-end logs
Fourth position	No supplied contribution evidence	Individual report and agreed responsibility record

A contribution declaration for each of the three documented members may be found in Appendix B. Since certification and agreement are human behaviours, signature fields are purposefully left empty. The missing fourth donation has not been confirmed or signed, according to the shared report.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common Technology Stack and Infrastructure

A shared application border and a mixed machine-learning environment are supported by the source set. A Python-oriented API design without inspect able architecture configurations and trained image-model roles are described in C1. ESP32, Firebase, and YOLO-based prototype components are reported by C2. PyTorch, torch vision, image preprocessing, and an exported classifier are all documented in C3 [20]. These materials are insufficient to construct a single verified mobile framework, production host, and shared database setup.

The suggested integration infrastructure and reported technology are separated in Table 5.1. This keeps an architecture diagram from being confused with a list of services that have been deployed.

Table 5.1 Technology stack and evidence status

Layer	Technology or design	Status in available evidence
Sensing	ESP32 environmental observations	Reported prototype in C2
Live telemetry	Firebase pathway	Reported prototype in C2
Stage vision	YOLOv8n, unspecified flowering YOLO, YOLO11s	Reported in C2; weights not inspected
Leaf screening	PyTorch and torchvision EfficientNet-B0	Documented C3 training and export contract
Land and ground vision	Segmentation and condition-analysis services	Roles documented; exact configurations absent
Shared API	Versioned Python or FastAPI-style services	Proposed design requiring repository reconciliation
Persistent records	Relational and spatial records; protected image storage	Logical target, deployment not verified
Mobile interface	Shared farmer-facing application	Prediction and guidance flows reported; framework unverified

The operational ramifications of the technology choices vary. Backend inference makes updating models easier, but it is dependent on hosting capacity and connectivity. Credential management and calibration upkeep are necessary for device

telemetry. Public picture databases add transfer and licensing restrictions but lessen the initial effort required for collecting. Rather than being concealed behind a list of technologies, these implications are taken into account in the deployment and commercialisation sections.

Open-source and enterprise license routes are documented by Ultralytics [30]. Before a commercial release, the project must examine the license of the precise software, models, and intended distribution. Open availability is not the same as unrestricted reuse in this study. Framework rights and dataset rights must be evaluated independently.

5.2 Development and Integration Process

5.2.1 Evidence-led development sequence

Source reconciliation is the first step in the standard procedure. Through optional capabilities, the four-sensor contract of C1 and the five-sensor contract of C2 are reconciled. YOLO detection is kept apart from C3's classifier. Model descriptions from the past are simply kept for context. These choices result in clear interface specifications that may be compared to the implementation.

Choosing a single repeatable baseline for every component is the next stage. Source revision, dependency data, model checksum, preprocessing, and a minimum input fixture make up the baseline. It would be challenging to replicate subsequent failures if a component were connected without first determining that baseline. It is impossible to determine whether the same weights and transformation were applied in a subsequent experiment using only a visual demonstration.

Short, reviewable modifications should be used in incremental integration. C1 and C2 owners assess a plan-to-cycle contract change, while C3 and the application maintainer review a classifier-output change. The review examines both successful output and field meaning and failure behaviour. For deterministic contract checks, continuous integration is advised; however, no pipeline is promised without configuration evidence.

5.2.2 Shared data preparation and model discipline

The most thorough preparation record is provided by the disease workflow. It chooses BBWV2/healthy photos, mobile-captured anthracnose material, selected raw COLD classes, and original rather than pre-augmented photographs from CHILLI [22], [23], [31], and [32]. Without suggesting that the provided PDFs contain all of the original downloads or their manifests, the group report maintains those sources.

The chosen class inventory reported before to duplication and conflict processing is replicated in Table 5.2. These are not held-out performance results; rather, they are preparation counts published by the source.

Table 5.2 Reported disease image inventory before filtering

Category	Selected images	Interpretation
Healthy	2197	Healthy category from selected sources
Bacterial spot	156	Source-labelled condition
Cercospora leaf spot	332	Source-labelled condition
Leaf curl	423	Source category includes curl-virus wording
Powdery mildew	102	Selected raw images
Anthracnose	162	Selected mobile-source images
BBWV2	1135	Controlled-environment source label
Total	4507	Before duplicate and conflict processing

After processing, 4,473 pictures remained, a net reduction of 34, according to the C3 report. It is impossible to reconstruct the precise retained class and split counts because the final manifest was not provided. Because augmented or near-duplicate views are not independent samples, it is crucial to group related images before partitioning. Plant-disjoint evaluation is also required for C2, and geographically isolated or otherwise leakage-aware splits are required for C1.

A configured 30-epoch schedule with three head-training epochs, seed 42, batch size 32, and roughly grouped 70/15/15 partitions are all documented in C3. AdamW is used, and validation macro-F1 is used to choose the optimal checkpoint. These are facts about configuration. They do not prove that the last test evaluation was carried

out or that all epochs were completed. The actual run log and the specified program must continue to be distinct records.

5.2.3 Checkpoints and deployment parity

Saving weights is not enough for interrupted training. The optimiser state, random state, dataset identification, and pertinent setup are all preserved in a resumable experiment. A fresh session shouldn't carry on with incompatible preprocessing or silently regenerate a different split. Despite the differences in their model formats, all components are subject to the same principle.

Particularly important is the C3 deployment contract. EfficientNet-B0 and its standard weights are documented by Torchvision [33], however the exported project model is based on the project's documented contain-and-pad transformation. The precise tensor generated from a non-square, orientation-tagged image in the training and serving pipelines is compared in a backend parity test. Only once this fundamental equivalency has been established should model correctness be examined.

A unique label map is required for the C2 detector bundle. While describing unripe and ripening semantics, the source report also notes that checkpoint names need to be verified. The confirmed mapping and visible counts from those classes should be made public by the service. Without verifying the trained labels, it must not infer that any class referred to be fruit is inherently unripe.

5.3 Deployment and Operational Configuration

5.3.1 Release package and environments

The application revision, schema versions, model bundles, rule profiles, and agriculture guide content should all be listed in a release. Different credentials and data boundaries should be used in development, test, and field-pilot settings. A real farmer's past must not be affected by test fixtures. Resource restrictions, permitted image formats, and the location of operating logs should all be specified in the configuration.

Startup tests confirm output dimensions, model availability, and compatible metadata. The API can provide a health status that separates each model's readiness from the availability of basic services. There is no inference function provided for a healthy

web process with missing weights. Information about readiness should be operationally helpful without disclosing internal file locations or secrets to regular users.

This research did not examine the database schema, instance size, deployment automation, or the real hosting provider. Thus, the deployment configuration to confirm and maintain is described in the requirements in this section. They are not proof that the cloud installation was finished.

5.3.2 Monitoring and recovery

Validated requests, denied inputs, timeouts, stale sensor windows, and unavailable-model events should all be counted in operational monitoring. These classifications separate infrastructure or model failures from user-input issues. An increase in ambiguous classifications might be the result of altered image conditions rather than server outages; the analysis should preserve sufficient context to distinguish between them.

Records, model metadata, rule versions, and references to protected assets should all be included in backups. An outdated recommendation may become irreproducible if merely a database is restored and the corresponding model version is lost. A short authorized test cycle should be restored, its evidence should be retrieved, and recovery exercises should verify that the same record linkages are still present.

Rollback ought to protect the past. If a new model fails validation, it can be taken out of active serving, but its prior outputs can still be attributed to it. The impacted version and the rationale for the modification should be noted in an agricultural advice correction. Subject to an established operational guideline, users should be notified appropriately when an earlier recommendation needs to be corrected.

5.3.3 Data retention and field operation

Research reuse and operational necessity should be distinguished in the retention policy. For every analysis, it is not required to retain every image forever. When photos are kept for validation, it should be obvious why they are there and how to access them. Unnecessary precise coordinates and personal identifiers should be avoided in derived anonymized evaluation records.

Practical device checks for placement, power, connectivity, calibration, and plot assignment are also necessary for field deployment. Recommendations from a sensor that is theoretically online but has been relocated may be deceptive. Therefore, reassignment must be clearly documented and cannot give the impression that historical observations came from the new plot.

5.4 Evidence of Team Execution

The provided member theses detail specific prototype actions and provide the established division of labor. Repository history, MS Planner exports, Teams meeting minutes, a release registration, and a whole research journal are not included. As a result, it is not possible to present sprint dates, commit totals, review participation, and completion percentages as verified group results.

The evidence required to support execution is listed in Table 5.3. This is not a replacement for the missing records; rather, it is an evidence registry.

Table 5.3 Team execution evidence register

Evidence class	Present in source set	What additional evidence would establish
Individual technical scope	Three member theses	Agreed final contribution boundaries
Prototype progress	Reported indirectly in C2 and C3	Original demonstration and release identifiers
Version control	Not supplied	Authorship, review and integration history
Task and meeting records	Not supplied	Planned work, decisions and dependencies
Release configuration	Not supplied	Exact build and deployed model set
Executed test records	Not supplied	Repeatable pass or fail outcomes
Field and expert review	Not supplied	Participation, findings and changes made

However, the current synthesis reveals particular integration work that the team can confirm: task-specific image processing, faithful classifier loading, plan-revision linkage, optional EC handling, and common failure conditions. These technical

decisions have acceptance criteria and can be reviewed. Compared to a generic claim that all modules are integrated, they offer a more solid foundation for following execution evidence.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation Framework and Success Criteria

6.1.1 Group evaluation questions

Four questions are separated by evaluation. Does every component make the right calculations or predictions? Do the interfaces maintain their meaning? Are the intended users able to comprehend and finish the workflow? Does utilising the workflow lead to better judgements or practical results? A positive response to one question cannot be used as a substitute for the others.

The faithful transmission of evidence is the fundamental integration criterion. An approved plan must arrive at the appropriate crop cycle while maintaining its constraints and version. The appropriate category under the appropriate model contract must be displayed in a leaf result. Confident reliant guidance cannot be produced by outdated or invalid readings. Every recommendation that is made should be able to be linked to the rule version and accompanying data.

Before assessing an independent test set, numerical prediction thresholds should be decided upon. This report does not create standards for yield improvement, mAP, or justified minimum accuracy because they are not established in the source documents. It is now possible to specify deterministic invariants: no cross-farm access, no class-index remapping, no zero-filled missing measurements, and no final suitability class in the absence of necessary proof.

6.1.2 Task-specific measures

Class intersection over union and area error against reference labels should be reported by segmentation evaluation. Bounding-box matching, class-level precision, recall, and average precision are necessary for object detection. A recognised detection-evaluation reference is offered by COCO's assessment resources [34]. A confusion matrix, per-class precision, recall, and F1 are necessary for leaf classification; macro-F1 reveals minority-class performance.

With the zero-denominator policy chosen, precision = TP/(TP + FP), recall = TP/(TP + FN), and F1 = 2PR/(P + R) for a class. The mean of the absolute difference between the predicted and annotated visible counts is known as the mean absolute count error.

It is necessary to measure sensor error using the same basis and temporal window as a suitable reference equipment. It is not possible to integrate these measurements in a meaningful way by averaging their percentages.

Deterministic unit verification, cost treatment, and cash timing are the first steps in financial computations. Then, a defined forecast horizon and observed results are needed for forecast validation. When reporting prediction intervals, it is important to look at both their empirical coverage and width. A wide range may encompass a variety of possibilities, making it useless for making farming decisions.

Evaluation layers are linked to evidence in Table 6.1. A documented version, dataset, participant foundation, and analytic process are required for every proposed study.

Table 6.1 Evaluation framework and evidence requirements

Layer	Principal measure or invariant	Evidence required
Land analysis	Per-class IoU and area error	Independent reference masks and geometry
Stage vision	Detection AP and visible-count error	Plant-disjoint annotated images
Leaf screening	Macro-F1, per-class recall and calibration	Fixed test split and prediction export
Sensors	Error, drift, missingness and freshness	Reference instruments and timestamps
Integration	Identity, unit and version preservation	Executed contract and end-to-end logs
Usability	Task success and limitation comprehension	Consented sessions and task records
Farming value	Decision quality, costs and observed outcomes	Prospective comparison with baseline practice

6.1.3 Baselines and study design

It is important to evaluate the integrated process with relevant alternatives. The same expert-reviewed cases can be used to evaluate image-only, sensor-only, and combined evidence for guidance. Arithmetic and explanations can be tested for planning using a manual calculation using the same inputs. If model selection is a research goal, a different pretrained backbone or a more straightforward classifier can be contrasted on the same fixed split for leaf screening.

When feasible, field evaluation should maintain independence between farms, plants, and seasons. A single plant captured in several photos is a correlated observation. It is not appropriate to present their number as the total number of independent agricultural situations. While higher outcome claims necessitate a design that takes weather, cultivar, management, and self-selection impacts into account, a pilot can detect usability and operational issues.

6.2 Integration and End to End Testing

6.2.1 Contract and regression tests

Priority integration tests are defined in Table 6.2. None of these cases are marked as passed without an execution record; they are specific acceptance cases. Expanded fixtures and the data needed to run them are provided in Appendix C.

Table 6.2 Priority integration tests and expected outcomes

ID	Scenario	Expected outcome
INT01	Accept plan revision and open crop cycle	Correct plot, count, method and revision retained
INT02	Use another farm's plan or image ID	Access denied without exposing protected content
INT03	Submit duplicate device message	One logical observation; retry acknowledged
INT04	Supply four-sensor device without EC	EC remains unavailable; unrelated guidance can continue
INT05	Supply stale or wrong-basis moisture	Dependent recommendation suppressed with reason
INT06	Send one invalid ground photo	Replace failed photo only; preserve accepted input
INT07	Return no detected fruit	Ripening fraction inestimable; no whole-plant claim
INT08	Exercise classifier indices zero to six	Exact mapping in Table 3.1 retained
INT09	Compare notebook and backend preprocessing	Equivalent oriented, padded and normalized tensor
INT10	Remove classifier weights	Unavailable state; no random-weight prediction

ID	Scenario	Expected outcome
INT11	Retry image request after timeout	Result remains tied to original image and request
INT12	Revise guidance content	Old recommendation retains original guidance version

The focus of regression cases should be on significant cross-component hazards. Tests for both new and old producers are required for a schema update. Count and label checks are required for a detector update. Category and uncertainty checks are necessary for a knowledge-base upgrade. Testing the altered boundary cannot be replaced by rerunning unrelated training.

6.2.2 End-to-end scenario

A defined plot and an approved test farm are the first steps in a complete scenario. The tester provides resource evidence, two ground images, and a legitimate overhead shot. Based on the completeness of the fixtures, C1 provides a temporary or suitable plan. After approving the plan, the tester initiates a crop cycle. A stage image is examined, a distinct leaf image is screened, and sensor messages are consumed.

The plan modification, crop cycle, observations, analysis, and recommendations with consistent ownership and timestamps are all included in the anticipated final history. A farmer acknowledgement is then recorded by the tester. The test confirms that later evidence does not supersede the initial plan and that an admission does not alter the prediction. Screenshots or response grabs, the underlying identifiers, and the release version are all included in a full run.

Failures such as expired evidence, an unavailable model, and an interrupted upload are injected in a second run. The system ought to display actionable recovery and preserve finished steps. This is crucial because perfect input or continuous connectivity cannot be assumed in the desired environment. Since there is no executed record of this entire scenario in the source set, its conclusion is still unknown.

6.2.3 Threshold and uncertainty checks

A top-two gap of 0.15 and a tentative top-score criterion of 0.70 are specified by C3 [3]. A top score of at least 0.70 leaves at most 0.30 for the second score, meaning the

gap is at least 0.40, since seven softmax scores add up to one. Therefore, among predictions that already satisfy the usual top-score criterion, the 0.15 rule adds no rejection. It only becomes relevant in certain contexts.

This is not a disease detection experiment; rather, it is an observation of mathematical consistency. Boundary equality and adjustable behavior should be tested without displaying the margin as additional protection at default settings. Unrelated objects, non-target leaves, and unrepresented symptoms must all be evaluated independently for unsupported images, and high-score failures and deferrals must be noted.

6.3 Overall Results

6.3.1 Supported findings

A cogent design with three distinct roles and documented prototype components is supported by the evidence that is now available. Although C1 provides a staged land and resource workflow, it does not provide held-out results or inspect able model configurations. Stage-specific detectors and the sensor/Firebase pathway are reported by C2. A documented preparation methodology and a mobile-connected classifier are reported by C3 [20]. Although they do not prove total system effectiveness, their results encourage more integration efforts.

The results at the strongest level supported by the source set are listed in Table 6.3. For reported prototype operation, documented configuration, and missing measurements, the language is purposefully different.

Table 6.3 Consolidated results and evidence strength

Item	Supported finding	Limit of conclusion
Component scope	Three member responsibilities identified	Fourth component not established
Planning design	Land, soil, resource and financial gates specified	Full implementation and predictive validity unverified
Guidance prototype	ESP32/Firebase and YOLO stages reported	No audited detector metrics or calibration series
Leaf preparation	4507 selected; 4473 retained reported	Final manifest not independently inspected

Item	Supported finding	Limit of conclusion
Leaf inference	Seven-category mobile prediction flow reported	Held-out accuracy and field performance absent
Advice layer	Source-based rule design documented	Completed agricultural review not supplied
Integrated operation	Shared contracts synthesized in this report	Complete executed end-to-end evidence absent
User and farm outcomes	Evaluation approach specified	No completed study or benefit estimate

The inventory maths is internally consistent: the reported retained total suggests 34 removals, and the seven chosen class counts add up to 4,507. Only the arithmetic replicated from C3 is validated by this check. It does not establish model quality, calculate retained class counts, or audit the photos.

6.3.2 Findings from cross-component synthesis

The primary integration requirement is data meaning, according to the initial synthesis finding. A disease checkpoint cannot be loaded as a YOLO detector based only on its filename; a fruit count cannot be handled as harvested mass; and a four-sensor plan cannot consume an assumed EC value. These mistakes can be identified through explicit contracts before they become recommendations that affect farmers.

The second conclusion is that there must be ongoing uncertainty between components. Even if a later sensor stream is valid, the crop-cycle record should maintain its status if the plan starts with provisional soil information. Placing a leaf image next to a reliable stage detector does not guarantee the health conclusion if the image is ambiguous. A well-designed combined dashboard does not prove that such doubts have been eliminated.

Research reporting is the subject of the third finding. A preparation inventory, not an accuracy or outcome result, is the source set's strongest quantitative detail. As a result, the report's integrative design and clear evaluation structure allow it to contribute in a way that can be justified. Until equivalent measurements are available, it cannot substantiate a claim of improved profit, decreased disease incidence, or increased yield.

6.4 Stakeholder and User Validation

There was no agricultural-expert validation record or full group-level participant study provided. As a result, participant counts, interview results, acceptance ratings, and recommendations are not disclosed. This document's intended advantages are specifications or theories that are awaiting assessment.

Growers should be asked to identify the approved plan, identify a stale sensor reading, provide a good leaf photograph, and explain an ambiguous outcome in a formative study. After documenting completion, assistance needed, and misunderstandings, the researcher should talk about the participant's justification. If a user confuses confidence for accuracy or thinks that a healthy-looking leaf indicates the entire field is disease-free, satisfaction on its own is insufficient.

Crop-reference applicability, sensor basis, stage norms, and management phrasing should all be examined by experts. Conflicting evidence and circumstances outside the classifier's purview should be included in cases. Reviewers should note whether the next course of action is suitable, whether causal claims are unsupported, and whether more investigation is required. Disagreements are not mistakes to be concealed by an unqualified consensus assertion, but rather results to be analysed.

Recruitment, consent, tasks, languages, analysis technique, and any ensuing design modifications should all be documented in the study record. The restricted generalisability of small pilot results should be acknowledged. Subsequent field research can evaluate whether users truly adhere to advice and whether doing so results in beneficial consequences in the local context.

6.5 Achievement of Common Objectives

The available evidence is mapped to the goals in Section 2.4 in Table 6.4. Completed implementation or field validation are not the same as documentation achievement.

Table 6.4 Objective achievement assessment

Objective	Status	Justification
G1 Common workflow	Achieved at design level	C1 to C3 relationships and boundaries documented
G2 Shared evidence contracts	Achieved at specification level	Identity, units, time and versions defined; execution pending
G3 Plan-to-monitoring integration	Partly substantiated	Producer and consumer designs exist; full run not supplied
G4 Faithful leaf screening	Partly substantiated	Prototype reported; parity and guidance tests remain
G5 Controlled failure handling	Specified but unverified	Acceptance cases defined; execution logs absent
G6 User and farming value	Not demonstrated	Held-out group evidence and prospective study absent

Documented design work is not diminished by the evaluation. It pinpoints the precise extra data that modifies each conclusion. A successful plan-to-cycle test, for instance, would bolster G3 without proving agronomic advantage under G6. Instead of rewriting the entire project as either complete or incomplete, maintaining these differences enables future modifications to update the pertinent claim.

6.6 Integrated Limitations and Threats

6.6.1 Internal and construct validity

When similar photos show up in various divisions, data leakage can exaggerate model estimates. The final manifest is required to validate the grouped splitting suggested by the disease workflow. Plant-level separation is necessary for detection research, and a suitable geographic divide is necessary for land studies. Software behavior is established by synthetic fixtures, but biological recognition cannot be measured by them.

Construct validity is concerned with the true meaning of a metric. Ripening proportion at the image level does not indicate complete harvest readiness. A thorough evaluation of soil suitability is not provided by soil sensor coverage. An accurate knowledge base search does not guarantee that the recommendations are suitable for the local context.

This is addressed by the evaluation system, which gives each claim a unique reference standard.

6.6.2 External and operational validity

Images of laboratory peppers and chillies from public sources may not match those from Sri Lankan Scotch Bonnet farms in terms of cultivar, background, lighting, disease presentation, and management. Instead of using disease features, a model can take advantage of source-specific features. An independently gathered local test set would come after per-source and leave-source-out analysis to help investigate this issue.

Network conditions, device placement, calibration drift, and user behaviour all affect operational validity. On hurried field captures, a detector that was tested on carefully chosen photos can behave differently. The suggested service model may underestimate the amount of maintenance required for a low-cost sensor. These elements must be supported by field data and ought to be factored into feasibility assessments.

6.6.3 Integration and evidence threats

The report is based on other documents that provide an overview of the underlying implementation data. There was no direct inspection of the original checkpoints, repositories, or PP2 artefacts. As a result, those documents are credited with reported progress. A full four-component evaluation is not possible due to the missing fourth report.

The test specs and architecture created in this synthesis are design contributions that can be reviewed. As historical implementation operations, they shouldn't be retroactive. The elements that were initially reported, those that were reconciled here, and those that were later carried out must be preserved in future iterations. The final research account's legitimacy is safeguarded by this division.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value Proposition and Beneficiaries

Continuity of decision support throughout planning, cultivation monitoring, and leaf screening is the suggested value. While getting task-specific interpretations, a grower can maintain one farm environment. Reduced effort in gathering and balancing data, together with more lucid justifications for missing evidence, are the possible benefits. There was no willingness-to-pay, customer conversion, or realized savings analysis provided.

Access to a well-organized case history is the suggested value for agricultural advisors. An adviser could examine the plot, crop stage, pertinent sensor base, and prior actions rather than just a symptom snapshot. A consultation may become more informed as a result, but its impact on advisory time or decision quality needs to be monitored.

At this point of evidence, the system should be positioned as research-based decision support. Both the supported scope and the available results would be exceeded by a commercial claim that it assures profit or identifies every pepper ailment.
Unrestricted deployment is not as defendable as a bounded supervised pilot.

7.2 Market Alternatives and Differentiation

Alternatives include independent planning spreadsheets, standalone sensor dashboards, general crop-image software, and manual farm records and adviser visits. Even without the complete NAIMIRIS methodology, each can offer benefit. Rather than a confirmed feature audit of identified items, the comparison in Table 7.1 focuses on methodology.

Table 7.1 Alternative approaches and proposed differentiation

Alternative	Practical strength	Proposed NAIMIRIS distinction
Farmer records and adviser visits	Local knowledge and contextual interpretation	Structured evidence history between visits

Alternative	Practical strength	Proposed NAIMIRIS distinction
Standalone sensor dashboard	Repeated environmental visibility	Link readings to stage and accepted plan
General image screening	Convenient symptom indication	Fixed crop-health contract within crop history
Planning spreadsheet	Transparent editable calculations	Connect assumptions to land and later observations
Separate specialist tools	Mature task-specific functions	Shared identity, provenance and recovery workflow

The integration design for the selected agricultural context is the defensible differentiation. It is neither a proven accuracy lead nor the creation of YOLO or Efficient Net. A dated checklist of supported crops, sensor requirements, offline behavior, advise review, and pricing should be used in any future competitor review. Limited marketing pages should not be used to infer claims about missing competition features.

7.3 Business Model and Feasibility

A supervised service provided by grower associations or agricultural advisers is a viable early model. Installation and hardware could be managed independently of application access. This is not an established relationship; rather, it is a suggested path to assess. Instead of supposing that software is enough on its own, it enables calibration support and user training to be taken into account as part of delivery.

Hosting, storage, picture processing, networking, device upkeep, calibration, support, and agricultural content evaluation are examples of potential ongoing expenses. For a small pilot, travel and hardware replacement can be substantial. Before deciding on a subscription or service fee, the group should measure consumption and get dated quotations. Here, no monetary estimate is made.

Explicit variables can be used to represent the provider's financial viability. When p surpasses v, the simplified break-even count is $F/(p - v)$, where p is the monthly income per active customer, v is the variable monthly cost, and F is the fixed monthly cost. The computation is merely a planning relationship; hardware expenses, seasonal consumption, turnover, and actual client acquisition must be handled separately. It is not to be confused with the ROI computation at the farmer level in C1.

Prototypes for mobile prediction and reported sensing provide some evidence of technical viability. Because service monitoring, recovery, calibration, and user assistance were not assessed in the source set, operational viability is still less certain. Evidence in each of these areas is necessary for commercial readiness. That evaluation includes more than just successful inference on a developer's machine.

7.4 Adoption Intellectual Property and Compliance

7.4.1 Adoption and ownership

The cost of the equipment, sporadic connectivity, the work required to take pictures, and the ambiguity around automated guidance are all obstacles to adoption. What each result signifies and how to fix weak evidence should be the main topics of instruction. When an optional sensor is not accessible, the application should still be able to make meaningful progress, withholding only the judgements that rely on it.

Ownership of models and code created by students should be balanced with agreements and institutional policies. Source imagery, framework code, and public datasets all have their own agreements. An asset registry that documents origin, version, license, and permissible use should be kept up to date by the research team. Commercial permission is not established by the presence of a downloadable dataset.

7.4.2 Data and software rights

The disease report documents licensing details for the sources it chooses, including limitations imposed on the CHILLI source [31]. Before using those phrases again, make sure they match the downloaded version. Different responsibilities may apply to derived model distribution and source selection. The group should not assert that the accessibility of its academic training procedure justifies the approval of a future commercial offering.

A particular evaluation of the selected deployment and distribution is also necessary for Ultralytics licensing [30]. Different serving arrangements can be accommodated by the integrated design, however architecture by itself cannot satisfy licence requirements. The actual software versions and review record should be preserved in a release decision.

7.4.3 Responsible advice and privacy

Images, financial assumptions, and perhaps sensitive farm locations are processed by the system. Prior to field collecting, appropriate privacy notifications, access controls, deletion plans, and institutional review should be set up. This thesis does not attest to compliance with applicable legal and regulatory requirements, which necessitate a current examination for the actual service and deployment jurisdiction.

A review procedure apart from model training is required for agricultural guidance. The knowledge base can be supported by source-based general sanitation and inspection data, but local crop, product, and label verification is necessary for chemical recommendations. The current design does not assert that vector control cures an infected plant and does not include automatic dose. Expert referrals should continue to be available for symptoms that are unclear or spreading.

Appendix E discloses the AI help used in the preparation of this common document. The disclosure separates experimental evidence from editorial synthesis. Before signing the statement, team members are still accountable for confirming their implementation claims, references, and final institutional requirements.

7.5 Sustainable Development Goal Contribution

Better decision support for small-scale farming is the planned route to SDG 2, while more thoughtful use of resources and agricultural inputs is the route to SDG 12 [8] Measurable indicators are required for these paths. Even though it presents itself as smart farming, a system that promotes the acquisition of needless equipment or excessive image-based interventions may result in expenses.

Without making any claims about observed progress, Table 7.2 lists potential markers. A predetermined baseline, time frame, and unit of comparison should be used in measurements.

Table 7.2 Sustainability pathways and evaluation indicators

Intended pathway	Possible indicator	Evidence and counter-risk
More feasible planting plans	Difference between planned and affordable count	Verified budgets; avoid overinvestment

Intended pathway	Possible indicator	Evidence and counter-risk
More informed water decisions	Water use per comparable production outcome	Calibrated measurement; avoid under-irrigation
More appropriate crop-health response	Expert-rated action appropriateness	Case review; avoid unnecessary treatment
Reduced information burden	Time and assistance needed per decision task	User study; account for device upkeep
Responsible device use	Repair, replacement and disposal record	Hardware lifecycle; avoid unmanaged waste

Benefits and costs would be assessed jointly in a realistic sustainability program. From the pilot stage, low-bandwidth operation, transparent advice, and device repairability can be taken into consideration. Only comparable outcome data should be used to support claims of increased revenue or decreased input utilisation.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall Conclusion

When farming decisions are made based on disparate observations, the common issue is the loss of context and evidence quality. The three NAIMIRIS components that are provided tackle complementary aspects of that issue: C1 organises land and investment planning, C2 connects crop development with environmental observations, and C3 uses a fixed seven-category classifier to filter leaf appearance. The faithful exchange of identity, time, units, versions, and uncertainty is what makes them valuable as a single system.

A defined integration design and reported prototype elements are supported by the evidence that is currently available. Complete end-to-end validation, field-level predictive performance, stakeholder acceptance, or financial gain are not established. The substantiated scope does not include the fourth component that the common template requires. Rather than being incidental constraints, these restrictions establish the conclusion's strength.

However, a traceable common workflow is a tangible result. It describes how sensor and image observations stay connected to the proper cycle, how an approved plan can direct crop registration, and how a health screening result can lead inspection without becoming a definitive diagnosis. Additionally, it outlines recovery in cases where evidence is outdated, insufficient, or unavailable. This makes it possible to test the subsequent stages of implementation and assessment.

8.2 Principal Integrated Contribution

An evidence-preserving integration definition for the three documented functions is the main contribution. It isolates model output from agronomic guidance, maintains task-specific model contracts, and resolves conflicting sensor assumptions through capability-aware records. It also establishes a link between the real maturity of the supporting material and objective assessment.

Beyond the specific models, a number of details are significant. Instead of passing a current plan that is subject to change, C1 must pass an approved revision. C2 needs to maintain the visible-image scope and measurement basis. C3 needs to replicate the

preprocessing and output order that were recorded. The evidence and rule versions that led to shared recommendations must be preserved. These specifications enable controlled future modifications and make the system inspectable.

Additionally, the report creates an assessment framework that prevents an unsubstantiated combined accuracy claim. Usability, interface accuracy, farming value, and predictive quality are all considered separate objectives. This allows the team to precisely show progress: an outcome claim requires a future farm study, whereas an integration claim is strengthened by passing a contract test.

8.3 Future Directions

8.3.1 Immediate evidence completion

Obtaining the final component artifacts—source revisions, model bundles, dataset manifests, calibration records, and executed test outputs—is the top priority. Only when its owner, scope, and supporting documentation have been verified should the fourth component be added. Instead of the existing three-report scope, this would enable the single report to reflect the entire group.

The team should then run the integration matrix and compare actual routes and storage with Appendix A. Early attention should be given to classifier loading and preprocessing parity because a single mismatch might render otherwise sound model work incorrect. Before understanding live advice as agronomically significant, sensor-based and stale-data instances should be carried out.

8.3.2 Local validation and prospective evaluation

To evaluate transfer from public sources, a locally gathered, expert-reviewed Scotch Bonnet dataset is required. In order for assessment to distinguish linked photos, it needs maintain farm, plant, season, and capture context. Because an apparently accurate closed-set model might nonetheless fail on inputs outside of its training scope, rare and unsupported circumstances are crucial.

The knowledge-base articles, stage profiles, and appropriateness guidelines should be examined by agricultural advisors. Understanding and practical application can then be evaluated by a formative farmer study. A prospective comparison should only look

at crop results, resource utilization, decision quality, and realized economics after those tests. Weather, cultivar, management, and participation variations must all be taken into consideration in the analysis.

8.3.3 Operational and product development

Operational development should prioritise reliable evidence capture, recovery and maintainability before expanding model scope. Useful work includes device calibration workflows, explicit offline states, versioned guidance and reproducible release packages. Multilingual presentation can improve accessibility if terminology and warnings are evaluated with intended users.

Additional capabilities such as severity estimation, automated actuation or expanded commercial services require their own datasets, contracts and evaluation. They should not be inferred from the existing classifier or added merely to fill a template position. The strongest path forward is to demonstrate the present workflow with independently reviewable evidence, then extend it where a clear user need and valid data justify the change.

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APPENDICES

Appendix A: Integrated Interface Specification

A.1 Status and scope

A standard interface for review and implementation is provided in this appendix. The routes are not validated deployed endpoints; rather, they are suggested logical contracts. Sections 4.1, 4.2, and 8.3 all make mention to it. If their documented semantics maintain the same ownership, input, output, and error requirements, the actual route names may vary.

Schema_version, request_id, farm_id, plot_id, crop_cycle_id where applicable, observed_at, recorded_at, evidence_ids, and quality status are all included in the common envelope. Actor and access are determined by the authenticated server context. A new picture or material input change generates a new analysis request; request identifiers are kept across safe retries.

Table A.1 Representative interface responsibilities

Operation	Proposed interface	Required validation
Create land analysis	POST /v1/land-analyses	Ownership, geometry, imagery and area basis
Create ground analysis	POST /v1/ground-analyses	Exactly two distinct usable images
Accept plan revision	POST /v1/plans/{id}/accept	Ownership, revision and visible limitations
Register crop cycle	POST /v1/crop-cycles	Accepted plan and valid plot relationship
Receive telemetry	POST /v1/observations/sensors	Device, sequence, units and calibration
Analyse stage image	POST /v1/analyses/stage-images	Cycle, stage and detector contract
Screen leaf image	POST /v1/analyses/leaf-images	Asset, model bundle and preprocessing
Read recommendation	GET /v1/recommendations/{id}	Farm access and immutable evidence links
Record action	POST /v1/recommendations/{id}/actions	Actor, action time and recommendation link

A.2 Plan and crop-cycle contract

Plan_id, revision, plot_revision, cultivation_method, feasible_plant_count, suitability_status, limiting_factors, missing_evidence, and source references are all included in a plan. Currency, season, price channel, cash-flow basis, and economic-cost basis are all identified in financial data. Corrections result in a new revision, but the accepted record is unchangeable.

Crop_cycle_id, plot_id, accepted_plan_id, accepted_plan_revision, cultivar when known, sowing or transplant date, current_stage, and stage provenance are all included in a crop cycle. The prior stage, actor, rationale, and effective time are all recorded in a rectified stage. The cycle does not imply an equal number of planned and surviving plants.

A.3 Telemetry dictionary

Only when a significant observation is present does each measurement store a numerical value. It is possible to keep raw readings and calibrated values apart. The system cannot deduce unavailable soil attributes from unrelated variables, and conversion rules document their version.

Table A.2 Telemetry fields and validation

Field	Meaning	Validation rule
message_id	Device event identifier	Unique within device identity
observed_at	Time of measurement	Explicit timezone; plausible clock relation
measurement	Named observed variable	Must be declared by device capabilities
value	Calibrated numerical reading	Finite number or null with reason
unit	Unit for the value	Compatible with measurement and profile
basis	Measurement interpretation	Required for moisture and EC comparison
calibration_id	Calibration record	Valid for device and relevant period
quality	Validity and freshness	Separate missing, stale and invalid states

A.4 Leaf-analysis response

Analysis_id, input_asset_id, model_version, preprocessing_version, label_map_version, class_index, class_key, seven scores, uncertainty_status, and guidance_version, if available, are all included in the response. Finite scores must be transferred to the precise order shown in Table 3.1. Instead of being referred to as measured accuracy, a confidence value is called a prediction score.

The service delivers a request reference and an unavailable status when weights are not available. Validation fails prior to inference when the image cannot be decoded. A filename or an earlier user's analysis is never used to rebuild a result. The classifier result may be accessible with an explicit guidance-unavailable state while advice is being reviewed.

A.5 Errors and compatibility

Validation errors reveal a field that can be fixed by the user without disclosing passwords or internal pathways. No protected image or financial material is revealed by authorisation failures. When appropriate, resource constraints result in a retryable response. Instead of discreetly removing a necessary unit or label-map field, schema incompatibility is expressed explicitly.

Additions that are backward compatible are still optional. Regression testing and a version change are necessary when class indices, moisture basis, or plan semantics are changed. A documented compatibility range should be maintained by both producers and consumers. In order to prevent future interpretation from relying on the present default, the request record maintains the version that was actually used.

Appendix B: Team Contribution Declaration

The three contributions shown in the individual reports provided are documented in this declaration. It does not certify an undocumented fourth contribution and supports Section 4.5. Only after reading the common report and the implementation evidence it references should the members verify the description and sign.

Pre-analysis and intelligent decision support, covering land and ground-image roles, evidence completeness, suitability, resources, feasible plant counts, and financial-scenario design, are covered in Kumarage K J B H, IT22271464.

Technology-assisted A-to-Z guidance, encompassing environmental monitoring, crop-stage workflow, stage-specific picture detection, and cultivation-advice design (Savinda R.M.D., IT22124630).

Gunawardana T.D.A., IT22212740: EfficientNet-B0 classification, stable class mapping, dataset preparation, and management-guidance design are all included in this AI-driven pest monitoring and control scope that is implemented as leaf screening.

Checking the shared interfaces, recording the actual application configuration, examining integration outcomes, and making sure that group-level assertions align with the evidence are all examples of shared responsibility. Without a confirmed record, authorship percentages, commit counts, and task dates are not allocated.

Table B.1 Member confirmation

Member and student ID	Signature	Date
Kumarage K J B H — IT22271464	_____	_____
Savinda R.M.D — IT22124630	_____	_____
Gunawardana T.D.A — IT22212740	_____	_____

Appendix C: Integrated Acceptance and Evaluation Protocol

C.1 Execution record

Case_id, build revision, schema version, model and rule versions, fixture identification, date, tester, actual output, expected output, result, and evidence location should all be included in each execution record. After repair, a failed case is still noted; a fresh run shows the fix. This appendix's examples are planned requirements rather than a record of tests that have been finished.

C.2 Identity and plan transfer

Prepare a single-plot farm with an approved plan change for INT01. From that revision, open a crop cycle and make changes to the editable plan. Its approved revision should still be shown by the crop cycle. The initial acceptance record, the cycle reaction, and the subsequent independent revision are anticipated pieces of evidence.

Make two authorised test users with different farms for INT02. Try to read or attach the plan, image, and recommendation IDs of the other user. Every protected action ought to be rejected. Even if the interface conceals certain identifiers, do the check at backend endpoints. Keep track of answers without disclosing actual user information.

C.3 Sensor capabilities and freshness

Submit a device message twice for INT03 and confirm that there is only one observation left. Later, provide an older buffered observation and confirm that ordering is determined by capture time.

Use a four-sensor capability record and leave out EC for INT04 and INT05. Next, submit two moisture values: one with an expired timestamp and the other with an unsupported basis. The program should differentiate between stale evidence, invalid bases, and inaccessible EC. A current legitimate crop condition or a zero should be used to denote none.

C.4 Image and model parity

Use two ground photos and one purposefully undecodable asset for INT06. While the unsuccessful photo is replaced, the approved photo must continue to be linked to the

request. In order to check that the ripening proportion is unestimable for INT07, inject a valid detector result that does not contain any fruit boxes.

Inject controlled classifier outputs for INT08, choosing each index from 0 to 6. Compare Table 3.1 with the key that is presented and stored. Use a portrait image with visible edge characteristics and EXIF rotation for INT09. Utilising a specified numerical tolerance, compare training and backend tensors. Instead than confirming disease accuracy, this confirms preprocessing parity.

Restart serving after removing the model from a test environment for INT10. The unavailable classifier should be identified by the readiness state, and requests shouldn't yield arbitrary predictions. Restore the authorised bundle and keep a separate record of the successful and unsuccessful cases.

C.5 Uncertainty fixtures

The default top-score deferral should be triggered by a synthetic distribution [0.69, 0.10, 0.07, 0.05, 0.04, 0.03, 0.02] that adds up to one. If the rule employs a stringent less-than comparison, a distribution [0.70, 0.10, 0.06, 0.05, 0.04, 0.03, 0.02] tests the precise threshold boundary. Only policy logic is verified by these fixtures.

Use [0.49, 0.40, 0.03, 0.03, 0.02, 0.02, 0.01] with a temporary top-score criterion of 0.40 and margin threshold of 0.15 to test the margin independently. Deferral should be triggered by the 0.09 gap. After the test, restore the documented configuration and keep both configurations in the log.

C.6 Outcome and usability evidence

The fixed test split, classes, grouping policy, and predictions are all recorded in the model evaluation. Usability documents tasks, consent, and miscommunications. Field evaluation documents observed results, management variations, and baseline practice. Every sort of evidence is evaluated independently.

Appendix D: Research Evidence and Reproducibility Register

D.1 Source authority

The authority for project-specific descriptions in this shared report is the three member reports. Concepts, procedures, datasets, and institutional guidelines are supported by external sources. The chapter structure is provided by the Common Report Content document, while the presenting guidelines are provided by Formating Standards. Only an example of a presentation is provided by the friend's group report.

Contradictory claims were settled cautiously. Instead of a YOLO pest detector, the present crop-health report establishes EfficientNet-B0. The advice paper does not specify the blooming variant and makes a distinction between seedling and fruit detectors. An inspectable final land-model architecture is not identified in the pre-analysis report. The common capability model preserves C1's absent EC.

D.2 Artefact register

The artefacts required to transform reported or specified work into repeatable evaluation are listed in Table D.1. Absence does not indicate that the team did not produce the item; rather, it indicates that it was not provided for this synthesis.

Table D.1 Reproducibility artefacts and current availability

Artefact	Purpose	Availability for this report
Three member theses	Component scope and reported progress	Supplied and analysed
Actual application repository	Verify routes, ownership and deployment	Not supplied
Land-model configuration and weights	Establish architecture and reproducibility	Not supplied
Stage detector bundles and label maps	Verify detector versions and counts	Not supplied
Leaf-model bundle and final manifest	Verify export and partitions	Described in C3; files not supplied
Training and test exports	Establish held-out metrics	Not supplied

Artefact	Purpose	Availability for this report
Calibration and telemetry logs	Establish measurement validity	Not supplied
Expert and participant records	Establish user and advice validation	Not supplied
Fourth individual report	Complete required group scope	Not supplied

D.3 Interpretation of reported quantities

A total of 4,473 photos were reportedly preserved out of the 4,507 selected photographs in the illness inventory prior to filtration. These numbers relate to the preparation of data. Since the estimated 70/15/15 split is a clustered process, it is unable to determine the precise number of images in each segment. A run record is needed for completed epochs, whereas a configured epoch count serves as a training plan.

This standard report does not include any graphs that show fake experimental results. The figures are unique logical diagrams created using the descriptions that were provided. They display state transitions, workflow, and architecture. Instead of being evaluated as recorded application output, they ought to be evaluated as design artefacts.

Appendix E: Academic Integrity and AI Assistance

The provided documents were analysed, contradictions were resolved, external sources were found, the common narrative was drafted and edited, integration specifications were created, and this Word document was formatted with the help of generative AI. Additionally, automatic aid was used to produce the diagrams and unexecuted test protocols. This disclosure does not prove that the mentioned system was developed or tested during the document preparation process.

The paper makes a distinction between freshly synthesised integrated design, configuration information, member-reported prototype progress, published research, and missing evaluation evidence. Research data was not replaced by participant responses, measurement accuracy, deployment uptime, yield gain, or profit growth.

The team is still in charge of verifying the dataset rights, implementation descriptions, institutional disclosure standards, and cited sources. The proper individuals must complete the supervisor certification and signatures. The relevant module policy and the actual development record should be followed when disclosing any extra AI use in code, data preparation, or analysis.